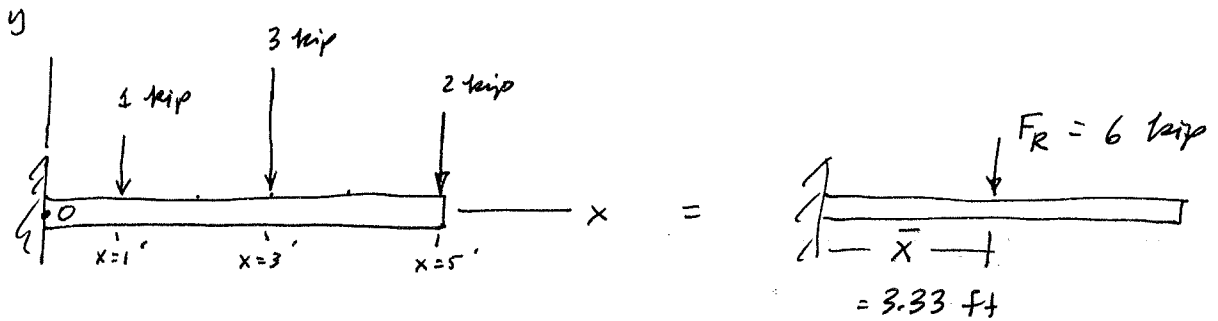


## Intro to Centroids



What if wanted to replace system of discrete forces w/  $F_R @ \bar{x}$ ?

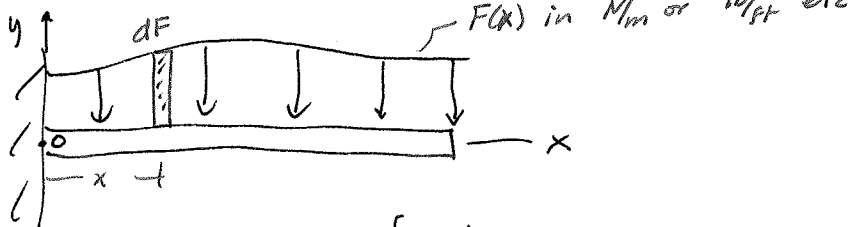
Resultant Force:  $F_R = \sum F_i$

Resultant Moment:  $\sum M_O = \sum F_i x_i = F_R \cdot \bar{x}$

$$\bar{x} = \frac{\sum F_i x_i}{F_R} = \frac{\sum F_i x_i}{\sum F_i}$$

$$= \frac{1(1) + 3(3) + 2(5)}{(1+3+2)} = \frac{20}{6} = \boxed{3.33 \text{ ft} = \bar{x}}$$

Distributed load



Resultant Force:  $F_R = \int dF = \int_A dA$  = Area under  $F$  curve

Resultant Moment:  $M_O = \int x dF =$  "first moment"  
 $= \int_A x dA$

$$\bar{x} = \frac{\int_A x dA}{\int_A dA}$$

Basically, a weighted average.